

DOVA

Local Development Runbook | August 2026

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Service: DOVA marketplace (NestJS API + Next.js storefront)

Environment: Localhost / developer machine

Last verified: 2026-08-29

App repo: [dreamcraft17/dova](https://github.com/dreamcraft17/dova)

Run the full DOVA stack on a laptop or workstation for feature work, QA, and debugging. Two modes:

Mode	When to use	Postgres	Redis
In-memory (default)	Fast UI demo, auth flows, cart/checkout mock pay	Not required	Not required
PostgreSQL	Migrations, seed data parity with prod, persistence	Required	Optional

Component	Dev command	URL
API (NestJS)	<code>`npm run dev -w apps/backend`</code>	<code>http://localhost:3000/api/v1</code>
Storefront (Next.js)	<code>`npm run dev -w apps/frontend`</code>	<code>http://localhost:3001</code>
Health	-	<code>http://localhost:3000/api/v1/health</code>

Both run together via root ``npm run dev``.

- **Node.js** 20+ and **npm** (monorepo workspaces)
- **Git** clone: ``git clone https://github.com/dreamcraft17/dova.git``
- **Ports free:** ``3000`` (API), ``3001`` (frontend dev)
- **Optional:** PostgreSQL 14+ and Redis 7+ (only if ``USE_IN_MEMORY=false``)

```
bash node -v # v20.x or v22.x npm -v lsof -i :3000 -i :3001 # should be empty before start
```

From repo root:

```
bash cd dova npm install cp .env.dev .env cp apps/backend/.env.dev apps/backend/.env cp apps/frontend/.env.dev apps/frontend/.env.local
```

Required env alignment (CORS)

``npm run dev`` serves the storefront on port 3001. Backend CORS must match:

```
env
```

```
FRONTEND_URL=http://localhost:3001 PAYSTACK_CALLBACK_URL=http://localhost:3001/checkout/verify
```

Root ``.env.dev`` already uses ``FRONTEND_URL=http://localhost:3001``. If registration or cookies fail with CORS errors, check this value first.

Path A - In-memory (recommended for first run)

No Docker, no Postgres.

```
bash cd dova npm run dev
```

Expected output

- Terminal shows ``BE`` and ``FE`` processes (concurrently)
- API log: ``DOVA API listening on :3000``
- Next.js: ``Local: http://localhost:3001``

Verify

```
bash curl -sf http://localhost:3000/api/v1/health
```

```
curl -sfl http://localhost:3001 | head -1
```

```
...
```

Open `http://localhost:3001` - catalog and demo login should load.

Path B - PostgreSQL (persistent data)

1. Create database:

```
bash createdb dova # or: psql -c "CREATE DATABASE dova;"
```

2. Set env:

```
```env
USE_IN_MEMORY=false DATABASE_URL=postgresql://postgres:postgres@localhost:5432/dova ```
```

## 3. Migrate and seed:

```
```bash npm run db:migrate npm run db:seed ```
```

4. Start:

```
```bash npm run dev ```
```

Verify: same health checks as Path A; login with demo accounts below.

1. In the terminal running `npm run dev`, press Ctrl+C once (stops both BE + FE). 2. Confirm ports released:

```
```bash lsof -i :3000 -i :3001 ```
```

3. If a process remains:

```
```bash kill $(lsof -t -i :3000) 2>/dev/null kill $(lsof -t -i :3001) 2>/dev/null ```
```

Check	Command	Pass
API up	`curl -sf http://localhost:3000/api/v1/health`	JSON with ok status
Frontend up	`curl -sfl http://localhost:3001 \`	head -1`
Auth (admin)	Login at `/auth/login` with demo admin	Redirect to `/admin`
Register OTP	Send code on `/auth/register`	Backend logs `[Reg OTP] email: #####`
Register success	Complete signup	Center Account created modal -> `/products`

Role	Email	Password
Admin	`admin@dova.local`	`admin1234`
Supplier	`supplier@dova.local`	`supplier1234`

Reset demo passwords (Postgres mode):

```
```bash npm run db:reset-logins ```
```

Email provider (Resend) is optional in development.

1. Open <http://localhost:3001/auth/register> 2. Enter email -> Send code 3. Read OTP from API terminal:

```
``` [Reg OTP] you@example.com: 123456 ```
```

4. Enter code + password -> Create account -> success modal

Production note: Resend (or SMTP) is required on VPS; see [ENV-SETUP.md](./ENV-SETUP.md).

QA fixed OTP (smoke): email `qa.softlaunch.N@example.com` + env `DOVA\_QA\_FIXED\_OTP=123456`.

`PAYSTACK_SECRET_KEY`	Behavior
Empty	Mock pay - checkout completes without Paystack UI
`sk_test_...`	Real Paystack test checkout - see [PAYSTACK-TEST-MODE.md](./PAYSTACK-TEST-MODE.md)

Frontend public key (optional for test UI): `NEXT\_PUBLIC\_PAYSTACK\_PUBLIC\_KEY=pk\_test\_...` in `apps/frontend/.env.local`.

```
```bash npm run dev          # API + frontend (watch mode) npm run build          # shared -> backend -> frontend npm run test:unit
    # Jest unit tests (shared + backend specs) npm run test          # unit + backend compile test npm run db:migrate    # apply SQL
migrations (Postgres) npm run db:seed          # demo catalog + accounts npm run smoke:week4    # local API smoke (set
API_URL=http://localhost:3000/api/v1) ```
```

Symptom	Likely cause	Fix
CORS error on login/register	`FRONTEND_URL` wrong port	Set `FRONTEND_URL=http://localhost:3001` in `apps/backend/.env`, restart dev
`EADDRINUSE :3000` or `:3001`	Previous dev server still running	Stop with Ctrl+C or `kill` (see Stop procedure)
Register: "Could not send verification email"	Only in production mode	Ensure `NODE_ENV=development` locally
OTP never arrives	No Resend in dev	Read `[Reg OTP]` line in backend terminal
Blank catalog	In-memory not seeded	Use demo login; or run `db:seed` in Postgres mode
`shared` import errors	Stale build	`npm run build -w shared` then `npm run dev`

Paystack redirect fails	Callback URL mismatch	<code>`PAYSTACK_CALLBACK_URL=http://localhost:3001/c heckout/verify`</code>
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In-memory mode: restart ``npm run dev`` - data resets on API restart.

Postgres mode:

```
``bash npm run db:migrate # idempotent npm run db:seed # refresh catalog + demo users npm run db:reset-logins ``
```

Nuclear (drop DB):

```
``bash dropdb dova && createdb dova npm run db:migrate && npm run db:seed ``
```

If a pull breaks dev:

```
``bash git stash git checkout main git pull npm ci npm run build -w shared cp .env.dev .env cp apps/backend/.env.dev  
apps/backend/.env cp apps/frontend/.env.dev apps/frontend/.env.local npm run dev ``
```

Restore stashed env if needed: ``git stash pop``.

Issue	Contact
Local env / runbook	Dozer

- [] Fresh clone + Path A start on macOS/Linux
- [] Register flow + modal success (Bug-016)
- [] Admin + supplier demo login
- [] Mock checkout completes
- [] ``npm run test:unit`` passes
- [] Update **Last verified** date above

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